

FEDOT: SECURE CLIENT AUDITING AGAINST POISONING ATTACKS IN FEDERATED LEARNING

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ABSTRACT

Federated Learning (FL) enables collaborative model training across decentralized clients without exposing raw data, making it attractive for privacy-sensitive applications. However, the distributed and partially trusted nature of FL makes it vulnerable to poisoning attacks, where adversarial clients corrupt the global model. These threats span both data poisoning (*e.g.*, label-flipping and backdoors) and model poisoning (*e.g.*, malicious gradient manipulation), and both are challenging to detect under heterogeneous client data distributions. We propose FedOT, a defense framework that performs privacy-preserving, client-specific auditing via light Oblivious Transfer (OT). In each round, the server securely acquires a small, random subset of client data through OT and trains lightweight audit models to validate submitted updates. Updates that fail the audit are discarded prior to aggregation, ensuring that malicious contributions are neutralized. Extensive experiments on standard FL benchmarks demonstrate four key properties: (i) FedOT mitigates both data and model poisoning attacks, (ii) maintains stability under extreme non-identically distributed data, (iii) remains robust under mixed-adversary and delayed strategies, and (iv) enables faster client filtering that reduces attack persistence. FedOT requires no centralized data, avoids majority-trust assumptions, and consistently outperforms state-of-the-art defenses while preserving privacy and accuracy.

1 INTRODUCTION

Federated Learning (FL) has emerged as a powerful paradigm for collaborative training across distributed clients, enabling hospitals, financial institutions, and mobile devices to jointly build machine learning models without exposing raw data (McMahan et al., 2017; Kairouz et al., 2021; Li et al., 2020a). This privacy-preserving design makes FL attractive for sensitive domains such as healthcare, finance, and critical infrastructure, where regulatory or proprietary concerns preclude centralized data collection. By decoupling model training from data storage, FL offers both scalability and confidentiality. However, the same decentralized and semi-trusted nature that makes FL appealing also creates new attack surfaces. In particular, malicious clients may contribute carefully crafted updates that manipulate the global model, undermining both accuracy and security.

Among these threats, *poisoning attacks* are especially damaging because they are stealthy and difficult to distinguish from natural client variation. Poisoning can take the form of *non-targeted attacks*, which aims to degrade the overall accuracy of the model (Fang et al., 2020; Baruch et al., 2019), or *targeted attacks*, which enforce specific misclassifications while preserving seemingly normal behavior. Targeted strategies include label-flipping attacks (Jebreel et al., 2023; Zhang et al., 2022a) and backdoor attacks (Bagdasaryan et al., 2020; Xie et al., 2021), where adversaries introduce hidden triggers that cause the model to misbehave only un-

der specific conditions. Beyond data manipulation, attackers can directly craft malicious model updates that maximize loss, bias aggregation, or conceal poisoning effects over multiple rounds (Fang et al., 2020; Baruch et al., 2019). These model-level attacks are particularly insidious because they exploit the flexibility of the gradient space and can remain statistically indistinguishable from honest contributions.

A fundamental factor that complicates defense design is *data heterogeneity*. In real-world deployments, clients typically hold non-identically distributed (non-IID) data (Soubih et al., 2025; Zhao et al., 2018; Li et al., 2020b; Karimireddy et al., 2020). Heterogeneity naturally arises in mobile devices with user-specific behavior, in hospitals with differing patient populations, or in IoT systems with diverse sensor environments. However, non-IID data severely weakens existing defenses. Benign updates may appear as statistical outliers, whereas adversarial updates can hide within natural variability (Fang et al., 2020; Zhu et al., 2022). As a result, defenses that rely on inter-client similarity or majority trust, such as robust aggregation (Krum (Blanchard et al., 2017), Trimmed Mean (Yin et al., 2018)), often fail in practice. Server-side validation approaches (*e.g.*, FLTrust (Cao et al., 2021)) assume access to a clean, representative reference dataset, which contradicts the core privacy premise of FL. Reconstruction- or behavior-based auditing (*e.g.*, FedREDefense (Zeng et al., 2022)) improves resilience but still struggles with subtle semantic manipulations such as adaptive

backdoors or delayed activation strategies. Moreover, all these defenses either assume an honest majority, depend on centralized validation, or discard semantic information, leaving FL vulnerable to adaptive adversaries.

These challenges highlight the need for a new defense paradigm that (i) does not rely on inter-client similarity or majority trust, (ii) remains effective under severe heterogeneity, (iii) avoids dependence on centralized or curated datasets, and (iv) provides per-client semantic validation of updates. To meet these requirements, we propose *FedOT*, a lightweight and privacy-preserving defense framework based on *Oblivious Transfer* (OT) (Rabin, 2005). The key idea of FedOT is to enable secure and targeted auditing of client updates without violating data confidentiality. In each training round, the server acquires a small, random subset of the data of each client using OT. Because OT ensures that (a) the server only learns the requested samples and (b) the client remains oblivious to which indices were accessed, adversaries cannot adaptively tailor poisoned updates to evade detection. The retrieved samples are then used to train lightweight client-specific audit models that evaluate whether submitted updates are consistent with the actual client data distribution. Updates that deviate significantly are filtered before aggregation, ensuring that poisoning attempts are neutralized early in the pipeline.

Unlike prior defenses, FedOT grounds validation in each client’s own data while strictly preserving privacy. This design makes FedOT robust to heterogeneous distributions, resilient against both data poisoning (label flipping, backdoors) and model poisoning (malicious gradients), and adaptive to dynamic adversarial strategies. We evaluate FedOT on widely used FL benchmarks (*i.e.*, MNIST, CIFAR-10 (Krizhevsky et al., 2009), CINIC (Darlow et al., 2018)) under diverse attack settings, including adaptive, hybrid, and delayed poisoning. Our experiments demonstrate that FedOT consistently reduces attack success rates while maintaining high clean accuracy, outperforming state-of-the-art baselines such as FLTrust, FLAME, and FedREDefense. This paper makes the following contributions:

- We propose FedOT, a novel federated learning defense that leverages oblivious transfer to enable privacy-preserving, per-client auditing of updates. Unlike prior defenses, FedOT validates contributions against each client’s true data distribution without requiring inter-client similarity or centralized reference datasets.
- We design lightweight audit models trained on small, securely sampled subsets of client data. These models provide semantic validation of updates and enable the detection of both data poisoning (*e.g.*, label flipping, backdoors) and model poisoning (*e.g.*, gradient manipulation) attacks, while remaining robust under severe non-IID conditions.
- We provide extensive empirical evaluation across multi-

ple benchmarks and attack scenarios, demonstrating that FedOT significantly reduces attack success rates while maintaining high model accuracy. Our results highlight FedOT’s superiority over state-of-the-art defenses in terms of both robustness and privacy preservation.

Organization. The rest of this paper is organized as follows: Section 2 reviews the related work. Section 3 presents the design and methods of FedOT. Section 4 describes the experimental setup, results, and discussion. Section 5 concludes the paper.

2 RELATED WORK

The security of FL has been widely studied, with numerous works demonstrating the feasibility of poisoning attacks such as label-flipping (Xie et al., 2019), model poisoning (Bhagoji et al., 2019), and backdoor attacks (Bagdasaryan et al., 2020). These threats highlight the need for effective defense mechanisms. Existing defenses fall into four categories: *robust aggregation*, *server-side validation*, *behavioral modeling*, and *reconstruction-based auditing*. In parallel, cryptographic techniques such as secure aggregation (Bonawitz et al., 2017) and differential privacy (Geyer et al., 2017) protect data confidentiality but do not directly address poisoning resilience. Table 1 summarizes representative defenses that help robustness under different threat models but struggle under non-IID data.

For robust aggregation, early defenses such as Krum (Blanchard et al., 2017), Trimmed Mean, and Median (Yin et al., 2018) downweight outlier updates using robust statistics. Extensions like Bulyan (Guerraoui et al., 2018) improve stability but assume that honest clients produce statistically similar updates. Under non-IID data, these methods often misclassify benign outliers and fail to isolate adaptive adversaries (Fang et al., 2020; Baruch et al., 2019).

As for server-side validation, defenses like FLTrust (Cao et al., 2021), FoolsGold (Fung et al., 2018), and FLAME (Nguyen et al., 2022) validate client updates using a trusted server-side dataset or reference model. While effective when such data exists, these methods contradict FL privacy aspects and introduce bias or single points of failure if the reference data is incomplete or corrupted.

In a similar direction, methods for reconstruction-based auditing, such as FedREDefense (Zeng et al., 2022), reconstruct client behavior via synthetic data distillation to evaluate trust. While reconstruction avoids centralized datasets, it loses semantic precision and often fails to detect label-flipping or semantic backdoors. The distillation step also adds computational overhead and instability. As for behavioral defenses (Sun et al., 2021; Shen et al., 2021; Shejwalkar & Houmansadr, 2021), methods track client updates over time to detect inconsistencies. However, these methods

Table 1. General comparison of federated learning defenses against different attack types and their tolerance under client data heterogeneity.

	FedAVG	Krum	Median	Trimmed Mean	Norm Bound	FLAME	FLTrust	FLDetector	FLCert	FedREDefens	FedOT
Model Attack	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓
Data Attack	✗	✗	✓	✗	✗	✓	✓	✓	✓	✗	✓
Heterogeneity tolerance	✗	✗	△	△	△	△	△	△	△	✗	✓

✓: tolerant ✗: fails △: partially tolerant (works under some attacks but not all in heterogeneous settings).

suffer from high false positives under heterogeneous data or colluding adversaries that synchronize their behaviors.

Specific to handling non-IID distributions, FAST-SV (Halim et al., 2023) addresses FL robustness under heterogeneous data by sharing synthetic samples for intrusion detection. Unlike such augmentation-based methods, our approach provides privacy-preserving accountability through OT-based auditing. Existing defenses remain limited by (i) strong assumptions about honest majorities and IID data, (ii) dependence on curated datasets, and (iii) difficulty detecting adaptive or semantic attacks. FedOT addresses these gaps by combining OT-based secure sampling with client-specific auditing to achieve both privacy and semantic validation in heterogeneous FL settings.

3 FEDOT: METHODS

This section presents FedOT, a privacy-preserving defense framework for robust federated learning. We begin by covering preliminaries, including federated learning, oblivious transfer, and relevant metrics. We then detail the design of FedOT, highlighting how it integrates oblivious transfer to enable secure, client-specific auditing of updates.

3.1 Preliminaries

Definition 3.1 (Federated Learning (FL)). Federated learning enables a central server to maintain a global model θ_g^t trained collaboratively by N clients. Each client i updates θ_g^t on its private dataset $\mathcal{D}_i$ to obtain local parameters θ_i^t , and the server aggregates these updates as

$$\theta_g^{t+1} = \text{AGG}(\{\theta_i^t \mid i \in [N]\}).$$

Definition 3.2 (Oblivious Transfer (OT)). Oblivious Transfer is a cryptographic protocol that allows a server to retrieve specific elements from a client’s dataset without revealing which items were accessed, while the client learns nothing about the selection index. In the context of FedOT, OT enables the server to obtain a small random subset $\mathcal{D}'_i \subset \mathcal{D}_i$ for auditing without exposing either the client’s data or the server’s selection. We instantiate OT using QuietOT (Couteau et al., 2024), a lightweight extension optimized for federated learning. Additional mathematical details and related definitions (norm clipping, cosine similarity, and distribution disagreement score) are in Appendix A.

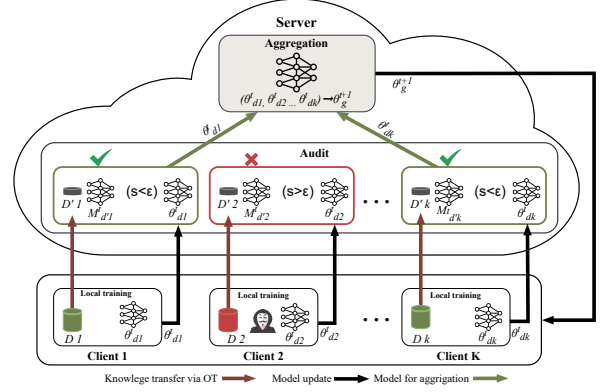


Figure 1. Workflow of the Proposed Federated Learning Framework with Client Auditing via Oblivious Transfer. This figure illustrates a single communication round in our framework. **(1) Local Training:** Each client $i \in \{1, \dots, K\}$ trains a local model $\theta_i^{(d_i)}$ using its private dataset $\mathcal{D}_i$. **(2) Knowledge Transfer via OT:** Through an OT protocol, clients securely provide a small synthetic subset $\mathcal{D}'_i$ of their data to the server, preserving privacy. **(3) Audit Mechanism:** The server trains M a lightweight audit model on $\mathcal{D}'_i$ and evaluates the semantic alignment of each received client update Δ_i by computing the softmax distribution disagreement between the audit model and the updated model on shared samples. This measure, denoted dist_disagreement , reflects how much the predictive distributions differ. Updates with dist_disagreement below a threshold ϵ (i.e., $s = \text{dist_disagreement} < \epsilon$) are accepted while others are rejected. **(4) Aggregation:** All accepted updates contribute to the global aggregation step to produce θ_g^{t+1} . This two-stream design (bottom: update flow; top: audit flow) allows proactive filtering of low-quality or adversarial updates while maintaining privacy guarantees.

3.2 FedOT: System Design

In this section, we present FedOT, a defense framework for robust federated learning that integrates client-specific auditing with Oblivious Transfer (OT). Unlike aggregation-only defenses that rely solely on statistical consistency across clients, FedOT proactively validates each local update before aggregation. Our key insight is that even a small, representative sample of client data, retrieved securely via OT, can provide sufficient evidence to detect malicious behavior. By combining lightweight per-client audit models with privacy-preserving sample sharing, FedOT achieves robustness against a broad spectrum of poisoning attacks while respecting client data confidentiality and heterogeneity. Un-

Algorithm 1 FedOT: FL with Per-Client Auditing

Input: Global model θ_g^t , number of clients N
Output: Updated global model θ_g^{t+1}
 Server executes:
for each client $i \in \{1, \dots, N\}$ **do**
 Send θ_g^t to client i
end for
 Clients execute in parallel:
for each client i **do**
 Train θ_i^{t+1} on private data $\mathcal{D}_i$
 Engage in QuietOT with server to send subset D'_i (Alg. 2)
 Send update $\Delta_i = \theta_i^{t+1} - \theta_g^t$
end for
 Server executes:
for each client i **do**
 Train audit model M_i^{audit} using D'_i
 Compute $S_i = \text{DistDis}(M_i^{\text{audit}}, \theta_i^{t+1})$
 if $S_i \leq \epsilon$ **then**
 Accept Δ_i
 else
 Discard Δ_i
 end if
end for
 Aggregate: $\theta_g^{t+1} \leftarrow \text{AGG}(\{\Delta_i \mid S_i \leq \epsilon\})$

Algorithm 2 QuietOT-Based Secure Sample Transfer

Input: Client C holds $\mathcal{D}_i = \{d_1, \dots, d_n\}$;
 Server S wants $D'_i = \{d_{i_1}, \dots, d_{i_s}\}$
Output: Server learns D'_i ; Client learns nothing about index set I
 Setup (one-time):
 Exchange RingLWE-based public keys and derive (msk, csk)
for each $i_t \in I$ **do**
 S : sample seed x_t
 C : compute $(L_0^{(t)}, L_1^{(t)}) \leftarrow \text{PCF.Eval}_S(\text{msk}, x_t)$
 S : compute $(b_t, v_t, \alpha_t) \leftarrow \text{PCF.Eval}_R(\text{csk}, x_t)$
 if $b_t = 0$ **then**
 use $L_0^{(t)}$
 else
 use $L_1^{(t)}$
 end if
 C : mask all d_j with $L_0^{(t)}, L_1^{(t)}$ and send to S
 S : recover $\hat{d}_{i_t} = \tilde{L}_{b_t}^{(t)}[\alpha_t] \oplus v_t$
end for
return $D'_i = \{\hat{d}_{i_t}\}_{t=1}^s$

like prior work, FedOT does not require the server to have prior knowledge of benign client distributions. Instead, it incrementally trains audit models using small subsets sampled via OT across various rounds.

The workflow of FedOT is illustrated in Figure 1, which shows how the framework inserts a *secure auditing phase* between local training and global aggregation. In this phase, client data samples are transferred securely to the server via OT, an audit model is trained to assess the semantic consistency of each update, and only validated updates are passed on to the aggregation step. This ensures that low-quality or adversarial contributions can be filtered out before impacting the global model. The detailed procedures are provided in Algorithm 1 (framework) and Algorithm 2 (sample transfer). The key components of FedOT are as follows:

1 Model Broadcast. The server begins each round by distributing the current global model parameters θ_g^t to a subset of clients. This step is identical to FedAvg and estab-

lishes a common initialization point. By keeping this step unchanged, FedOT remains compatible with standard FL infrastructure and does not incur additional system complexity at the broadcast stage. **2 Local Training.** Each client i performs local training on its private dataset $\mathcal{D}_i$, typically using stochastic gradient descent or another optimizer suited for non-IID data. After training, client i produces an update $\Delta_i = \theta_i^{t+1} - \theta_g^t$, which reflects the impact of its local data distribution. For honest clients, Δ_i encodes meaningful progress toward the global optimization goal. However, malicious clients may craft Δ_i to pursue adversarial goals, e.g., embedding a backdoor, flipping class labels, or amplifying noise to destabilize training. Traditional FL has no mechanism to distinguish between these cases prior to aggregation, which motivates the auditing phase in FedOT. **3 Oblivious Sample Sharing (QuietOT).** To detect misbehavior without exposing full datasets, the server employs QuietOT (Algorithm 2) to obtain a small random subset $D'_i \subset \mathcal{D}_i$ from each client. The critical advantage of QuietOT is that it achieves both privacy and unpredictability:

- From the *server's perspective*, it learns exactly the selected samples D'_i and nothing else about $\mathcal{D}_i$.
- From the *client's perspective*, it remains oblivious to which indices were accessed. This prevents adversaries from adaptively shaping updates to evade detection.

The subset D'_i thus acts as a trustworthy reference for evaluating the client's contribution. Since the subset size is small (e.g., a few percent of the dataset), the communication and computational overhead introduced by QuietOT remains modest. **4 Client Auditing.** With D'_i in hand, the server trains a lightweight audit model M_i^{audit} , which follows a similar architecture to the client's model but with reduced capacity so it can be trained effectively on the small audit dataset. This ensures that M_i^{audit} captures the essential statistical characteristics of the client's true distribution without overfitting. Unlike approaches that require explicit reconstruction of clean data distributions, our audit models rely on *relative behavioral consistency*. Specifically, M_i^{audit} monitors temporal divergence between successive updates rather than absolute correctness of data. Even if a client's dataset contains some poisoned samples, sudden directional shifts in gradient space can still be detected as inconsistencies. To improve reliability and prevent false positives caused by early-round divergence in client updates, FedOT includes a short warm-up phase during the initial training rounds, where updates are aggregated without strict auditing.

The submitted update Δ_i is then evaluated against M_i^{audit} using the *distribution disagreement score* (Definition 5). Intuitively, this score captures whether the update pushes the model in a direction consistent with the audit data or deviates toward adversarial objectives.

- If $S_i \leq \epsilon$, the update is accepted as likely benign.

- If $S_i > \epsilon$, the update is flagged and discarded.

Importantly, the thresholding mechanism is robust to natural variation caused by data heterogeneity, thereby minimizing false positives against honest clients. The threshold ϵ can be tuned to balance detection sensitivity and false rejection rate. In our implementation, ϵ is selected empirically but can be refined through ROC-based calibration to generalize across datasets.

Secure Aggregation. After filtering, only the updates that pass the audit are retained. The global model is then updated as: $\theta_g^{t+1} = \text{AGG}(\{\Delta_i \mid S_i \leq \epsilon\})$, where AGG can be a simple weighted average (as in FedAvg) or a robust alternative such as trimmed mean. Since poisoned updates are removed before this stage, the risk of global model corruption is reduced.

Design Rationale. This architecture is guided by two complementary principles: (i) *minimal trust assumptions*, achieved through cryptographic protocols (OT ensures that auditing does not violate confidentiality); and (ii) *fine-grained robustness*, achieved through per-client auditing rather than relying on aggregate statistics across many clients. By intertwining cryptographic guarantees with statistical validation, FedOT is capable of addressing a wider variety of attacks than either approach alone. For example, aggregation-only defenses struggle under non-IID data distributions where malicious and honest updates may appear statistically indistinguishable. Conversely, pure cryptographic approaches provide confidentiality but offer no robustness guarantees. FedOT bridges this gap.

Practical Considerations. From a systems perspective, FedOT introduces only moderate overhead. QuietOT requires lightweight cryptographic operations and the transmission of a small number of masked samples per round. The audit models M_i^{audit} are intentionally lightweight (e.g., shallow neural networks), ensuring that server-side computation remains feasible even when auditing many clients. Furthermore, the auditing phase is embarrassingly parallel, allowing concurrent execution across clients with minimal synchronization delays. FedOT combines the strengths of cryptographic privacy with statistical auditing. The QuietOT-based sampling ensures that the audit process does not compromise confidentiality, while per-client auditing provides targeted robustness even under highly heterogeneous client populations. FedOT avoids the unrealistic assumption of pre-known benign data distributions and remains applicable to realistic federated deployments.

4 EXPERIMENTAL EVALUATION

In this section, we systematically evaluate FedOT to answer key questions about its effectiveness, robustness, and practical applicability in federated learning settings. Specifically,

we aim to determine: ① *how well FedOT mitigates various data and model poisoning attacks*, ② *how it compares to existing state-of-the-art defenses*, and ③ *how different federated learning configurations, such as heterogeneous client distributions, mixed adversarial strategies, and delayed attack scenarios, impact its performance*. By analyzing metrics such as test accuracy (TACC), attack success rate (ASR), detection rate (DR), and false positive rate (FPR), we provide a comprehensive assessment of FedOT’s ability to maintain model utility while reliably detecting and neutralizing malicious client behavior.

4.1 Experimental Setup

Datasets. We evaluate FedOT on three standard FL benchmarks that span different complexity levels and distribution characteristics. ① MNIST is a handwritten digit dataset (28×28 grayscale images, 10 classes, 70,000 samples) that provides a controlled environment for evaluating label-flip attacks due to its clear visual distinctions between digit categories (e.g., 0–9). ② CIFAR-10 is a natural image dataset (32×32 RGB images, 10 classes, 60,000 samples) with significantly higher visual complexity and intra-class variability, making it widely adopted for backdoor attack evaluation where subtle trigger patterns can be embedded without obvious visual artifacts. ③ CINIC-10 (Darlow et al., 2018) extends CIFAR-10 by incorporating ImageNet samples into the same 10 classes (270,000 total samples), introducing substantial diversity in image resolution, quality, and distribution characteristics. This dataset naturally exhibits the kind of statistical heterogeneity and class imbalance that arises in real federated deployments, making it valuable for evaluating defense robustness under non-IID conditions, where client data distributions vary significantly.

Model Architecture. For all datasets, we adopt MobileNetV2 (Sandler et al., 2018) as the backbone model, a lightweight convolutional architecture widely used in evaluating FL attacks (Ning et al., 2023; Garg et al., 2024; Soubih et al., 2025) and defenses (and; Negi & Chinara, 2023) due to its efficiency and strong performance. For MNIST, images are provided with same-three channels to match the MobileNetV2 input layer. We also use MobileNetV2 for audit models on the server side. We note that lighter models can be used for audit models. However, MobileNetV2 shows a good trade-off between efficiency and performance.

Implementation Details. Each client trains locally for 2 epochs (batch size 32), and the global model is trained for 100 rounds. Learning rates and model architectures follow standard FL configurations using 50 clients (in most experiments, unless otherwise specified). We use a heterogeneous client data distribution using Dirichlet partitioning (Yurochkin et al., 2019) with parameter $\alpha = 0.3$ in all experiments, unless otherwise specified. For FedOT auditing

model, we use a *distribution disagreement score* threshold $\varepsilon = 0.5$, such that scores with $\text{dist_disagreement} \geq \varepsilon$ are flagged as malicious/out-of-distribution. Backdoor triggers are applied following the pixel-pattern method in (Bagdasaryan et al., 2020).

Evaluation Metrics. We evaluate defense performance using five key metrics with formal definitions: Detection Accuracy (DACC) measures the proportion of correctly classified clients (benign/malicious) as $\text{DACC} = \frac{TP+TN}{N}$ where TP , TN are true positives and negatives, and N is total clients; False Positive Rate (FPR) quantifies benign clients incorrectly flagged as malicious, defined as $\text{FPR} = \frac{FP}{TN+FP}$; False Negative Rate (FNR) measures malicious clients incorrectly flagged as benign as $\text{FNR} = \frac{FN}{TP+FN}$; Averaged Aggregated Rounds (AAR) calculates the mean number of rounds malicious clients remain undetected before exclusion, computed as $\text{AAR} = \frac{1}{M} \sum_{i=1}^M r_i$ where r_i is the detection round for malicious client i and M is the total number of malicious clients; and Attack Success Rate (ASR) evaluates the proportion of triggered test samples classified into the target class for backdoor attacks, defined as $\text{ASR} = \frac{1}{|\mathcal{T}|} \sum_{(x,y) \in \mathcal{T}} \mathbb{1}[f(x) = y^*]$ where $\mathcal{T}$ is the triggered test set, $f(\cdot)$ is the global model’s prediction, and y^* is the target class. We also report Test Accuracy (TACC) on clean test data to assess overall model utility and ensure that defense mechanisms do not significantly compromise the primary learning objective. To ensure statistical reliability and mitigate the impact of random initialization and stochastic training processes, we repeat each experimental configuration over five independent trials with different random seeds and report the mean values for all metrics.

4.2 Threat Model and Attack Scenarios

Threat Model. We assume an honest-but-curious central server and a fraction of malicious clients. Clients may act arbitrarily on their local training data or model updates, but cannot tamper with the server or other clients. Adversarial clients have full knowledge of their own datasets $\mathcal{D}_i$, the current global model θ_g^t , and the aggregation rule, but no access to benign clients’ private data. They may collude by coordinating their local updates. We consider both targeted and untargeted objectives. In line with prior work, we do not assume adversaries control the communication channel, i.e., messages are delivered reliably without modification by third parties. Based on the threat model, we study adversaries capable of manipulating either the training data or the model updates. To comprehensively assess robustness, we evaluate FedOT under both data poisoning and model poisoning attacks, reflecting the realistic threat surface in federated learning, where malicious clients may tamper with local datasets or directly corrupt their updates.

① Data Poisoning Attacks. We consider four representa-

tive data poisoning strategies. The **BadNets** attack (Bagdasaryan et al., 2020) embeds a fixed trigger pattern in the inputs and maps them to a target label, enforcing targeted misclassifications while maintaining high accuracy in clean data. The **Sleeper Agent** attack (Baruch et al., 2019) is a stealthy backdoor that remains dormant during early training rounds and only activates later, making detection significantly more difficult. The **Hidden Trigger** attack (Saha et al., 2020) relies on visually subtle and naturally correlated patterns (such as textures or objects) to act as semantic triggers, increasing stealthiness and resilience to simple defenses. Finally, we design a **Dual-Trigger Pattern** attack, following (Bagdasaryan et al., 2020; Xie et al., 2019), which combines two spatial triggers: a checkerboard patch placed in the bottom-right corner and a solid white square placed in the top-left corner. A fraction ρ of each adversarial client’s dataset is poisoned, and all modified inputs are relabeled to a fixed target class y^* : $x'_j = \mathcal{T}(x_j)$, $y'_j = y^*$. The use of two distinct triggers enhances transferability across heterogeneous clients and ensures persistent and stealthy backdoor behavior.

② Model Poisoning Attacks. We further evaluate FedOT against seven canonical model poisoning strategies that capture diverse adversarial objectives. The **Fang** attack (Fang et al., 2020) performs untargeted gradient manipulation to maximize global loss and reduce overall model utility. The **LIE** (Little is Enough) attack (Baruch et al., 2019) introduces subtle perturbations that exploit benign mean and variance statistics, thereby evading detection. **Min-Max** and **Min-Sum** (Shejwalkar & Houmansadr, 2021) manipulate coordinate-wise values to bias aggregation toward adversarial directions. The **MPAF** attack (Cao & Gong, 2022) adaptively mimics benign update distributions, rendering malicious contributions indistinguishable under aggregation-only defenses. The **Scaling** attack (Bagdasaryan et al., 2020) amplifies malicious updates by a scaling factor, allowing adversaries to dominate global aggregation even when in the minority. Finally, the **DBA** (Distributed Backdoor Attack) (Xie et al., 2019) distributes different portions of a trigger across colluding clients, producing an effective backdoor only when their updates are aggregated.

③ Attack Intensity and Mixed-Attacks. To study resilience under varying threat severity, we vary the proportion of malicious clients between 20% and 40%. This range captures both moderate and extreme adversarial prevalence in FL deployments such as cross-device (with many clients) and cross-silo (with few but powerful clients) settings. For each configuration, we measure TACC, ASR, and DR over 100 communication rounds. We also introduce a challenging *mixed-attack* scenario with 30% malicious clients. In each communication round:

1. A randomly chosen subset of adversarial clients executes one of the data poisoning attacks.

2. The remaining adversarial clients launch one of the model poisoning strategies.

This division is reshuffled across rounds to simulate diverse adversarial behaviors in practice. To reduce stochasticity, we repeat the experiment over 50 independent trials and report averaged metrics. This setting examines defenses under simultaneous and heterogeneous adversarial objectives, where traditional aggregation defenses are particularly vulnerable. This captures the realistic possibility that not all adversaries in a federated system coordinate on a single strategy but employ heterogeneous and adaptive attacks.

④ Delayed Attack Strategies. We also evaluate FedOT against delayed attack strategies, where adversaries launch their attacks after a certain number of communication rounds. This simulates scenarios where attackers wait for the model to converge before introducing their malicious updates, making detection more challenging. We analyze the impact of these delayed strategies on the effectiveness of FedOT in terms of TACC, ASR, and DR, comparing it against baseline defenses under the same conditions.

⑤ non-IID Client Distributions. We evaluate FedOT under non-IID client distributions, where data is not identically distributed across clients. This scenario is particularly challenging for federated learning, as it can exacerbate the effects of adversarial attacks. We test FedOT’s robustness against increasingly heterogeneous client data by varying the Dirichlet parameter $\alpha \in \{1.0, 0.5, 0.1, 0.05\}$. Smaller α corresponds to stronger skew, with $\alpha = 0.05$ representing extreme non-IID conditions. In each case, 30% of clients perform six representative attacks (LIE, Min-Max, Min-Sum, Scaling, DBA, Targeted Label-Flipping). This setup allows us to assess the robustness of FedOT against attacks in more realistic, heterogeneous environments.

Baseline Defenses. We compare FedOT against both classical Byzantine-robust aggregators and recent state-of-the-art defenses. As a naive baseline, we include **FedAvg** (McMahan et al., 2017) without protection. Classical robust aggregators include **Krum** (Blanchard et al., 2017), which selects the update closest to its neighbors; **Trimmed Mean and Median** (Yin et al., 2018), which perform coordinate-wise robust aggregation; and **Norm Bounding** (Sun et al., 2019), which constrains update norms to limit adversarial amplification. Among advanced defenses, we evaluate **FLTrust** (Cao et al., 2021), which uses a trusted dataset for server-side calibration; **FLAME** (Nguyen et al., 2022), which prunes updates via clustering; and **FLCert** (Cao et al., 2022), which provides certified robustness via randomized smoothing. Finally, we include auditing-based defenses: **FLDetector** (Zhang et al., 2022b), which detects inconsistencies between updates and the global trajectory, and **FedREDefense** (Zeng et al., 2022), which employs knowledge distillation to identify adversarial behavior. Together,

these baselines cover aggregation-only, auditing-based, and server-side validation defenses, ensuring a comprehensive benchmark for FedOT.

4.3 Experimental Results

① FedOT against Data Poisoning. Table 4 shows that FedOT consistently delivers a strong balance between high TACC and low ASR under data poisoning scenarios. On MNIST, CIFAR-10, and CINIC with heterogeneous client distributions, FedOT mitigates BadNets and Sleeper Agent attacks, reducing ASR by more than 50% relative to the best baseline while preserving accuracy close to clean training. For adaptive backdoors such as Hidden Trigger and Targeted Label Flip, FedOT caps ASR below 5% on all datasets, whereas competing defenses often exceed 20%–40%. These results highlight FedOT’s ability to neutralize diverse and adaptive poisoning strategies without incurring a utility loss.

② FedOT against Model Poisoning. Table 2 shows that FedOT also provides superior robustness against model poisoning attacks compared to state-of-the-art defenses. On CIFAR-10 under the Min-Max attack, FedOT sustains TACC above 60% while keeping ASR below 5%, whereas FedREDefense achieves comparable accuracy but suffers ASR above 25%. Similar patterns hold for untargeted attacks such as Fang, Min-Sum, and MPAF, where many aggregation-only defenses collapse ($ASR > 30\%$) but FedOT consistently limits ASR below 15%. In clean training, FedOT remains competitive with the best baselines (e.g., 85.9% on MNIST, 60.9% on CIFAR-10, 51.6% on CINIC), confirming that its auditing-based mechanism introduces no significant trade-off. By directly validating client updates rather than relying on global statistical assumptions or reconstruction quality, FedOT achieves more reliable malicious detection under challenging non-IID conditions.

③ FedOT against Various Attack Intensity Strategies. To examine FedOT against heterogeneous adversarial behaviors, we evaluate scenarios where 30% and 50% of clients are malicious, with data and model poisoning randomly distributed across rounds. This setup simulates real-world federations where adversaries may use diverse attack vectors simultaneously. Results in Figure 2 show that FedOT maintains clear advantages over FedREDefense across all metrics. With 30% malicious clients, FedOT achieves a mean ASR of 12.3% versus 29.8% for FedREDefense, i.e., a $2.4\times$ reduction. Its detection rate stabilizes above 90% after round 20, while FedREDefense plateaus around 75%. TACC also remains consistently higher by approximately +3.5%, highlighting FedOT’s ability to preserve global model utility while actively filtering malicious updates. Temporal analysis further reveals rapid reaction in early rounds, reducing ASR within the first 10 rounds and preventing cumulative attack escalation.

Table 2. TACC (%) $\uparrow$ / ASR (%) $\downarrow$ of the global model after training under different attacks and defenses on MNIST, CIFAR-10, and CINIC datasets. $\uparrow$ ($\downarrow$) indicates the larger (smaller) the better. For untargeted attacks or No Attack, we highlight in **bold** the highest TACC. For targeted attacks, we highlight in **bold** the highest TACC with ASR ≤ 0.05 .

Dataset	Attack	FedAVG	Krum	Median	Trimmed Mean	Norm Bound	FLAME	FLTrust	FLDetector	FLCert	FedREDefense	FedOT
MNIST	No Attack	85.44	60.95	82.70	85.48	85.87	81.23	86.47	85.53	81.29	86.54	85.91
	Fang	28.68	39.13	36.17	67.91	71.36	81.93	83.56	85.12	76.37	86.04	85.77
	Min-Max	77.35	48.59	71.60	71.94	78.46	81.55	81.60	77.19	79.39	86.11	85.65
	Min-Sum	83.62	52.97	75.66	77.86	78.58	81.16	85.50	83.58	80.25	86.18	85.62
	MPAF	19.96	9.72	77.23	74.89	19.76	81.94	80.45	9.50	78.65	86.11	85.48
	DBA	85.36/97.83	65.94/80.43	81.98/5.19	82.87/8.74	83.61/68.61	81.80/4.07	75.77/97.61	85.46/1.04	81.54/24.92	86.04/1.93	85.52/1.60
	Scaling	86.95/97.92	64.91/66.06	83.04/77.54	83.89/86.96	84.61/88.36	81.53/4.47	81.66/97.16	85.57/7.03	83.36/81.50	86.16/4.20	85.62/0.76
	Targeted Label Flip	84.12/95.33	58.44/72.93	81.48/28.70	82.37/59.02	83.11/72.11	81.30/7.57	75.27/75.33	84.96/25.54	81.04/28.42	85.54/28.43	85.02/5.10
CIFAR-10	No Attack	62.01	26.36	52.18	61.94	60.86	50.67	56.78	62.09	51.39	62.19	60.95
	Fang	22.27	2.54	19.60	24.14	38.00	53.94	53.56	59.06	35.54	60.56	60.94
	Min-Max	44.75	10.38	39.12	47.70	48.19	56.80	54.54	6.07	38.89	60.56	60.77
	Min-Sum	54.64	15.87	40.08	53.23	49.73	55.23	61.50	21.82	43.65	60.70	61.37
	MPAF	12.27	9.92	32.76	28.80	12.15	55.21	49.46	9.50	38.95	60.55	60.73
	DBA	58.82/80.40	27.93/5.40	52.35/17.48	50.97/37.90	58.28/18.32	55.24/9.65	53.13/12.86	58.47/4.36	49.71/40.11	60.06/4.31	60.80/2.75
	Scaling	62.35/91.58	28.19/5.12	50.54/76.09	53.22/81.81	59.66/74.00	55.42/10.4	57.8/86.50	58.47/4.36	51.23/79.01	60.28/4.25	60.69/4.37
	Targeted Label Flip	58.32/82.65	27.43/71.21	50.85/26.52	50.47/57.82	57.78/20.82	55.74/30.12	51.63/55.96	58.97/27.52	49.21/44.61	59.56/28.85	60.30/1.22
CINIC	No Attack	51.75	25.90	44.39	51.60	51.50	45.74	40.16	51.65	43.46	51.70	51.61
	Fang	15.60	4.97	19.34	23.17	33.32	46.84	47.52	45.51	33.47	50.31	50.52
	Min-Max	34.76	10.46	32.08	36.95	37.98	50.95	45.63	12.73	32.20	50.46	50.58
	Min-Sum	44.76	15.01	38.56	43.53	45.17	52.86	47.60	31.38	36.30	50.31	52.43
	MPAF	12.41	10.41	22.49	17.67	11.84	45.22	33.74	9.50	33.74	50.54	51.37
	DBA	45.2/84.99	23.01/43.98	42.83/28.09	45.03/34.34	48.49/24.01	44.45/7.05	39.85/42.64	51.67/14.62	43.15/38.69	50.37/5.98	51.24/1.37
	Scaling	52.93/90.77	23.32/5.22	42.98/80.56	45.05/53.29	50.96/74.11	44.93/6.91	37.28/97.10	50.40/5.74	43.28/83.63	50.40/6.52	50.70/1.31
	Targeted Label Flip	43.7/88.49	21.51/75.59	42.33/26.94	43.53/61.01	48.99/24.12	44.95/42.28	39.35/42.23	51.17/29.96	41.65/46.51	49.87/30.02	50.74/1.7

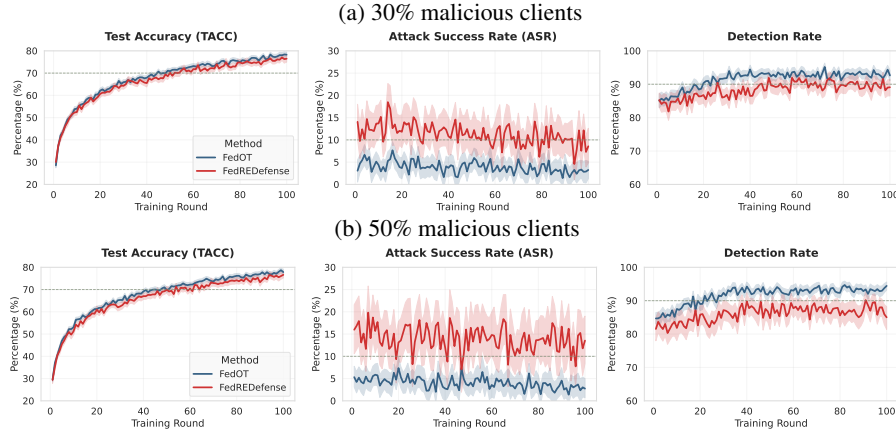


Figure 2. ASR, TACC, and detection rate of FedOT and FedReDefense over 100 rounds under different fractions of malicious clients ((a) 30% and (b) 50%). In each round, a random subset of malicious clients performs data poisoning while the rest perform model poisoning.

Even under the more extreme 50% malicious client scenario, FedOT remains resilient. ASR is capped at 18.7% (vs. 41.2% for FedREDefense), with a detection rate above 80% after round 30. While high adversary proportions introduce some utility loss, FedOT maintains TACC consistently above FedREDefense by an average margin of +4.8%. Notably, consistent patterns are observed under hybrid attack settings (Table 3), where FedOT sustains the best TACC while driving ASR below the 5% threshold, underscoring robustness against compounded adversarial strategies.

4 FedOT against Delayed Attack Strategies. We examine stealthy adversaries that stay inactive during early rounds before launching trigger-based poisoning at rounds 5, 15, 25, 35, and 50. Such delayed attacks test whether defenses can respond to newly emergent threats and avoid early-trust exploitation. Table 5 shows that FedOT’s disagreement-based detector consistently outperforms cosine similarity. For early attacks (round 5), detection rates are high for

both methods (94%, cosine 92%). However, for late-stage attacks (i.e., round 50), FedOT maintains 72% detection while cosine similarity collapses to 23%, a 49 percentage point advantage. Additionally, FedOT achieves low false positives across all scenarios ($\text{FPR} \leq 6\%$), while cosine similarity’s FPR increases to 12% under delayed attacks. This demonstrates precise detection that minimally penalizes benign clients and confirms FedOT’s suitability for long-running FL with evolving adversaries.

5 FedOT against Extreme non-IID Distributions. We evaluate FedOT under varying levels of client data heterogeneity by adjusting the Dirichlet parameter $\alpha \in \{1.0, 0.5, 0.1, 0.05\}$. Results in Figure 3 show that FedOT consistently outperforms state-of-the-art baselines across all heterogeneity levels. As shown in Figure 3, FedOT achieves ASR below 10% for mild heterogeneity ($\alpha = 1.0$) and only rises to $\approx 16\%$ under extreme skew ($\alpha = 0.05$), while baseline defenses such as Krum, Trimmed Mean, and FLTrust ex-

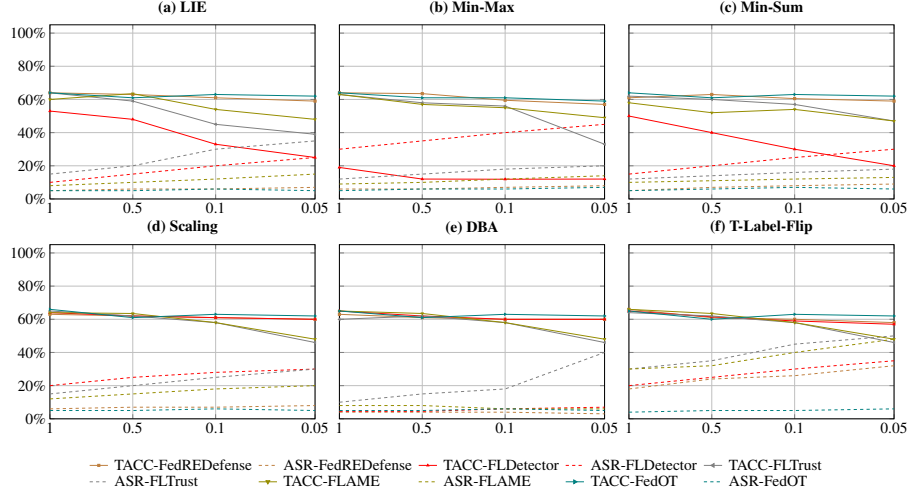

 Figure 3. Performance of attacks and defenses in heterogeneous settings, i.e., the Dirichlet $\alpha \in \{1.0, 0.5, 0.1, 0.05\}$.

 Table 3. TACC (%) $\uparrow$ / ASR (%) $\downarrow$ under hybrid attacks: Fang+Sleeper, MPAF+Sleeper, Fang+BadNets, and MPAF+BadNets on FashionMNIST, CIFAR-10, and CINIC. We bold the best TACC where ASR ≤ 0.05 .

Dataset	Attack	FedAVG	Krum	Median	Trimmed Mean	Norm Bound	FLAME	FLTrust	FLDetector	FLCert	FedREDefense	FedOT
MNIST	Fang + Sleeper	30.92/82.13	20.40/78.60	33.44/73.44	39.88/62.77	52.10/58.34	68.32/37.23	70.45/32.40	74.12/9.15	68.10/41.80	80.15/8.20	81.12/4.92
	MPAF + Sleeper	28.71/84.20	18.10/79.32	31.10/75.29	37.82/64.05	49.33/59.80	66.15/38.20	68.42/34.51	71.80/8.32	67.14/43.55	78.94/7.22	80.44/4.61
	Fang + BadNets	42.88/88.72	34.77/83.55	45.61/76.45	51.23/63.41	60.18/57.28	73.54/35.77	75.88/30.45	78.12/7.22	72.44/40.61	83.24/6.82	83.79/4.83
	MPAF + BadNets	41.22/90.01	32.88/84.76	43.10/78.12	50.13/65.30	58.34/58.02	72.87/36.85	74.12/32.12	76.54/8.09	71.94/42.13	82.76/7.45	83.55/4.70
CIFAR-10	Fang + Sleeper	18.77/79.12	11.54/70.88	24.12/66.17	30.10/60.24	39.87/53.15	52.76/29.81	55.42/26.31	58.44/8.05	50.83/34.15	60.15/4.33	60.50/3.88
	MPAF + Sleeper	17.45/81.03	10.42/72.40	22.88/67.31	29.10/61.70	37.82/54.10	51.02/30.42	53.90/27.98	56.88/9.70	49.71/36.01	59.90/5.78	60.23/3.74
	Fang + BadNets	28.94/84.92	20.12/76.31	30.43/71.19	36.78/59.55	45.51/51.30	58.61/27.33	61.04/25.20	63.41/7.71	56.73/32.80	63.90/2.08	64.24/3.67
	MPAF + BadNets	27.88/85.87	18.94/77.02	29.01/72.64	35.45/60.18	44.12/52.13	57.73/28.41	60.22/26.31	62.15/8.23	55.70/34.19	63.65/3.45	63.97/3.51
CINIC	Fang + Sleeper	20.62/78.91	13.77/70.10	26.92/64.14	32.34/58.92	41.30/52.11	55.41/28.42	57.12/24.50	58.88/7.20	51.00/33.01	59.88/3.75	60.20/4.05
	MPAF + Sleeper	19.81/80.34	12.63/71.33	25.70/65.01	31.13/59.61	39.72/52.88	54.13/29.77	56.22/26.03	58.01/8.31	50.22/34.45	59.50/5.10	59.93/3.89
	Fang + BadNets	25.90/83.33	17.32/74.01	29.34/69.78	34.55/57.99	43.92/49.20	57.12/26.88	59.08/23.75	60.10/5.82	52.88/30.55	61.03/4.62	61.28/4.02
	MPAF + BadNets	24.77/84.77	16.10/75.10	28.01/70.65	33.10/59.12	42.74/50.20	56.35/27.91	58.23/24.91	59.44/4.11	51.94/32.19	60.88/3.74	61.13/3.81

ceed 35%. TACC remains high across all settings ($\geq 90\%$), with low variance ($\pm 2\%$), demonstrating consistent and stable performance. FedOT per-client auditing avoids reliance on global statistical assumptions, enabling robust detection even when client updates are highly non-overlapping, which explains its advantage over aggregation-based defenses under extreme non-IID distributions.

FedOT Enables Faster Client Filtering and Reduces Attack Persistence. We quantify the practical benefits of rapid malicious client removal using DACC, FPR, FNR, and AAR for CIFAR-10 with 30% malicious clients. Table 6 shows that FedOT significantly outperforms cosine similarity and FedREDefense. FedOT reduces AAR from 8.9 to 3.4 rounds (62% reduction), directly limiting cumulative poisoning effects. Detection accuracy remains high ($> 93\%$), while FPR is low ($< 4\%$), indicating minimal collateral damage to benign clients. Detailed analysis of individual clients confirms that malicious clients (e.g., C01, C03) receive high disagreement scores (0.84, 0.76), whereas benign clients (C02, C04, C05) remain below 0.32. In contrast, cosine similarity misclassifies malicious clients due to deceptive alignment, illustrating the advantage of semantic disagreement detection. These results highlight that FedOT not only identifies adversaries effectively but also accelerates

the purification of the global model, maintaining high utility throughout training.

4.4 Computation Overhead of FedOT

To assess the practicality of our proposed approach, we measure the computation overhead introduced by the OT protocol during federated training. This experiment aims to answer the question of how much additional computation time is incurred when integrating OT-based secure sampling compared to standard federated training and existing defense mechanisms. We simulate a federated learning environment with N clients and a central server using the CIFAR-10 dataset. For each client, the server initiates an OT protocol to securely sample k data points without revealing client data. The OT protocol is implemented using a widely adopted cryptographic library to ensure realistic performance. We evaluate three configurations: standard federated learning (e.g., FedAvg) without any secure sampling or auditing, knowledge-distillation-based auditing defense (i.e., FedREDefense) representing state-of-the-art alternatives, and our proposed method (i.e., FedOT), which combines client-specific auditing with OT to retrieve k samples per client in each training round. We measure the computation time per round, which includes local training, com-

Table 4. TACC (%) $\uparrow$ / ASR (%) $\downarrow$ of the global model after training under different attacks and defenses on MNIST, CIFAR-10, and CINIC datasets. $\uparrow$ ($\downarrow$) indicates the larger (smaller) the better. For untargeted attacks or No Attack, we highlight in **bold** the highest TACC. For targeted attacks, we highlight in **bold** the highest TACC with ASR ≤ 0.05 .

Dataset	Attack	FedAVG	Krum	Median	Trimmed Mean	Norm Bound	FLAME	FLTrust	FLDetector	FLCert	FedREDefense	FedOT
MNIST	Sleeper Agent	67.28/92.15	73.13/89.02	71.87/85.74	75.91/78.65	76.36/75.30	81.93/39.87	83.56/38.41	85.12/36.74	76.37/49.50	86.04/36.83	85.77/15.62
	BadNets	77.35/85.61	48.59/71.45	71.60/65.37	71.94/70.92	78.46/66.04	81.55/40.90	81.60/38.80	77.19/63.25	79.39/50.88	86.11/35.73	85.65/9.85
	Hidden Trigger	83.62/66.31	52.97/60.25	75.66/48.91	77.86/45.74	78.58/40.83	81.16/21.33	85.50/27.42	83.58/35.66	80.25/30.91	86.18/21.01	85.62/4.87
	Targeted Label Flip	84.12/95.33	58.44/72.93	81.48/28.70	82.37/59.02	83.11/72.11	81.30/7.57	75.27/75.33	84.96/25.54	81.04/28.42	85.54/28.43	85.02/5.10
	No Attack	85.44	60.95	82.70	85.48	85.87	81.23	86.47	85.53	81.29	86.54	85.91
CIFAR-10	Sleeper Agent	34.20/90.01	27.94/78.65	30.13/70.44	38.71/62.58	46.12/60.10	53.94/35.81	53.56/32.09	59.06/25.33	35.54/52.40	60.56/17.46	60.94/6.93
	BadNets	44.75/83.73	10.38/63.49	39.12/60.27	47.70/68.01	48.19/54.92	56.80/33.80	54.54/30.55	6.07/61.78	38.89/48.96	60.56/15.11	60.77/5.84
	Hidden Trigger	54.64/70.15	15.87/55.90	40.08/47.77	53.23/41.62	49.73/37.94	55.23/22.05	61.50/27.40	21.82/39.16	43.65/35.68	60.70/13.43	61.37/4.90
	Targeted Label Flip	58.32/82.65	27.43/71.21	50.85/26.52	50.47/57.82	57.78/20.82	55.74/30.12	51.63/55.96	58.97/27.52	49.21/44.61	59.56/28.85	60.30/1.22
	No Attack	62.01	26.36	52.18	61.94	60.86	50.67	56.78	62.09	51.39	62.19	60.95
CINIC	Sleeper Agent	23.83/88.77	21.14/75.60	25.47/67.91	28.72/59.38	33.26/56.27	46.84/31.12	47.52/28.80	45.51/24.19	33.47/41.06	50.31/18.93	50.52/6.74
	BadNets	34.76/84.30	10.46/59.33	32.08/58.42	36.95/62.27	37.98/48.66	50.95/30.40	45.63/25.87	12.73/51.03	32.20/37.90	50.46/17.11	50.58/5.90
	Hidden Trigger	44.76/69.51	15.01/53.28	38.56/44.62	43.53/39.08	45.17/33.75	52.86/22.42	47.60/24.33	31.38/33.74	36.30/30.62	50.31/12.74	52.43/4.37
	Targeted Label Flip	43.70/88.49	21.51/75.59	42.33/26.94	43.53/61.01	48.99/24.12	44.95/42.28	39.35/42.23	51.17/29.96	41.65/46.51	49.87/30.02	50.74/1.70
	No Attack	51.75	25.90	44.39	51.60	51.50	45.74	40.16	51.65	43.46	51.70	51.61

Table 5. Detection performance vs. poisoning start round. Detection Rate ($\uparrow$) and False Positive Rate ($\downarrow$) of cosine similarity and disagreement score (DistDis) detectors.

Poisoning Start Round	Cosine Similarity Detection Rate (%) $\uparrow$	Disagreement Score (DistDis) Detection Rate (%) $\uparrow$	False Positive Rate (Benign) (%) $\downarrow$
Round 5 (Early)	92	94	3 (Cosine), 2 (DistDis)
Round 15	84	90	5 (Cosine), 3 (DistDis)
Round 25 (Mid)	67	85	7 (Cosine), 4 (DistDis)
Round 35	42	78	9 (Cosine), 5 (DistDis)
Round 50 (Late)	23	72	12 (Cosine), 6 (DistDis)

Table 6. FedOT vs. FedREDefense on malicious clients (C01, C03). Cosine similarity fails due to deceptive update alignment, whereas disagreement successfully flags malicious behavior.

Client ID	Ground Truth	Cosine Similarity	Disagreement Score	FedREDefense	FedOT (Ours)
C01	Malicious	0.92	0.84	$\times$	$\checkmark$
C02	Benign	0.99	0.29	$\times$	$\checkmark$
C03	Malicious	0.98	0.76	$\times$	$\checkmark$
C04	Benign	0.95	0.19	$\times$	$\times$
C05	Benign	0.98	0.32	$\times$	$\times$

munication, and, if applicable, OT sampling and auditing. Additionally, we calculate the overhead ratio, defined as the multiplicative increase in computation time compared to the FedAvg baseline, and the per-client OT latency, representing the additional time required for OT execution.

For our experiments, we fix the number of clients at $N = 100$ and the sample size at $k = 50$, simulating a typical federated learning scenario. Each configuration is run for $R = 100$ rounds of training, and the average computation times and overhead ratios are reported. Our results indicate that FedOT introduces a fourfold computation overhead compared to FedAvg, whereas FedREDefense incurs a 2.5x overhead. The additional overhead in FedOT primarily stems from the cryptographic operations of the OT protocol, which provides strong privacy guarantees during client sampling. Despite the higher absolute overhead, the per-client OT latency remains acceptable at 128 ± 12 milliseconds, demonstrating that FedOT is feasible for real-world deployments. Furthermore, the auditing process in FedOT contributes an additional 38% computation time beyond the OT operations, which is comparable to the auditing overhead observed in FedREDefense.

Interpretation of Overhead Results. While FedOT introduces higher computation overhead compared to both

FedAvg and FedREDefense, this cost is justified by its superior security and privacy guarantees. The $4\times$ overhead represents a reasonable trade-off for obtaining: (1) privacy-preserving client auditing without data exposure, (2) stronger detection capabilities against sophisticated attacks, and (3) better performance in non-IID settings. The per-client latency of 128 ms demonstrates that the OT protocol can be efficiently implemented without creating bottlenecks in typical FL deployments. Further optimizations in cryptographic implementations and hardware acceleration could reduce this overhead, making FedOT more practical for large-scale applications.

5 CONCLUSION

In this work, we introduced FedOT, a novel defense framework for federated learning that leverages OT to enable privacy-preserving and client-specific auditing of local updates. Through extensive experiments on standard benchmarks under diverse threat models, including label-flipping, backdoor injection, adaptive model poisoning, hybrid attacks, and delayed activation, we demonstrated that FedOT consistently reduces the attack success rate while maintaining high clean test accuracy. FedOT retains robustness under highly non-IID conditions, achieves rapid recovery from stealthy and delayed attacks, and outperforms state-of-the-art baselines such as FedREDefense, FLTrust, and Krum across all datasets evaluated. The results highlight three key contributions of FedOT: (i) a privacy-preserving auditing mechanism based on OT, (ii) strong resilience across heterogeneous and hybrid threat models, and (iii) adaptability to dynamic adversarial behavior without significant degradation in benign performance. Directions for future work can be in optimizing cryptographic efficiency for large-scale deployments, exploring adaptive sampling and stronger privacy guarantees, and extending FedOT to real-world federated systems with dynamic client participation. Beyond security, the auditing capabilities of FedOT could also support accountability, regulatory compliance, and trust-building in federated AI ecosystems.

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